Regulatory Sandbox: Minor Change Approval in Support of a Major Modification

AEA Canada Regional

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Overview

Understanding Regulatory Sandboxes

– What is a Regulatory Sandbox?

Our Regulatory Sandbox in Focus

– Minor Change Approvals: Background & Objectives

– What is Being Varied: Overview of Regulatory Adjustments

– How it will Work: Exemption, Advisory and Guidance Material

Questions

Outline

Today, we will highlight content from the following materials:

- Exemption NCR-028-2023
  - Grants relief from CARs 521.152(2), 571.06(1)(a), and 571.10(1).
  - Is the regulatory tool that enables the regulatory sandbox for minor change approvals.
- Advisory Circular (AC) 521-XXX (Draft) – Issuance of a Minor Change Approval Used in Support of a Major Modification
  - Information and guidance on expectations for aircraft certification delegates and AMO's/AME's.
- Staff Instruction (SI) 500-28 – Regulatory Sandbox: Minor Change Approvals – Administration, Delegate Selection and Delegate Authorization Process
  - Guidance to Transport Canada aircraft certification staff on administration of the regulatory sandbox.

"Success is the sum of small efforts, repeated day in and day out."

Robert Collier

Introduction

Issue:

- Transport Canada is considering amendments to the *Canadian Aviation Regulations* (CARs) to allow aircraft certification delegates to approve third-party (non-TC/STC holder) engineering technical data for eligible major modifications without requiring a Supplemental Type Certificate (STC).

Regulatory Sandbox:

- The sandbox will allow Transport Canada to test and demonstrate, in a controlled environment, a new method for approving engineering technical data for simple major modifications.
- Evidence gathered will inform potential amendments to the CARs.
- Transport Canada is looking for volunteer participants.

What are Regulatory Sandboxes?

Temporary Limited Authorizations:

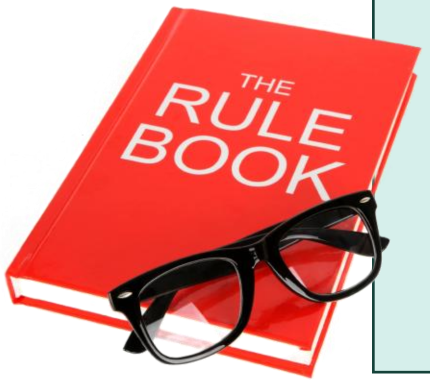
- Allow for temporary limited authorizations of innovation under regulatory supervision.

Learning and Modernization:

- Help regulators learn how to incorporate or best regulate innovation and demonstrate how regulatory regimes could be modernized.

Evidence-Based Decisions:

- Evidence gathered can help regulators decide on permanent changes, including management of innovative products, services, processes, and approaches.

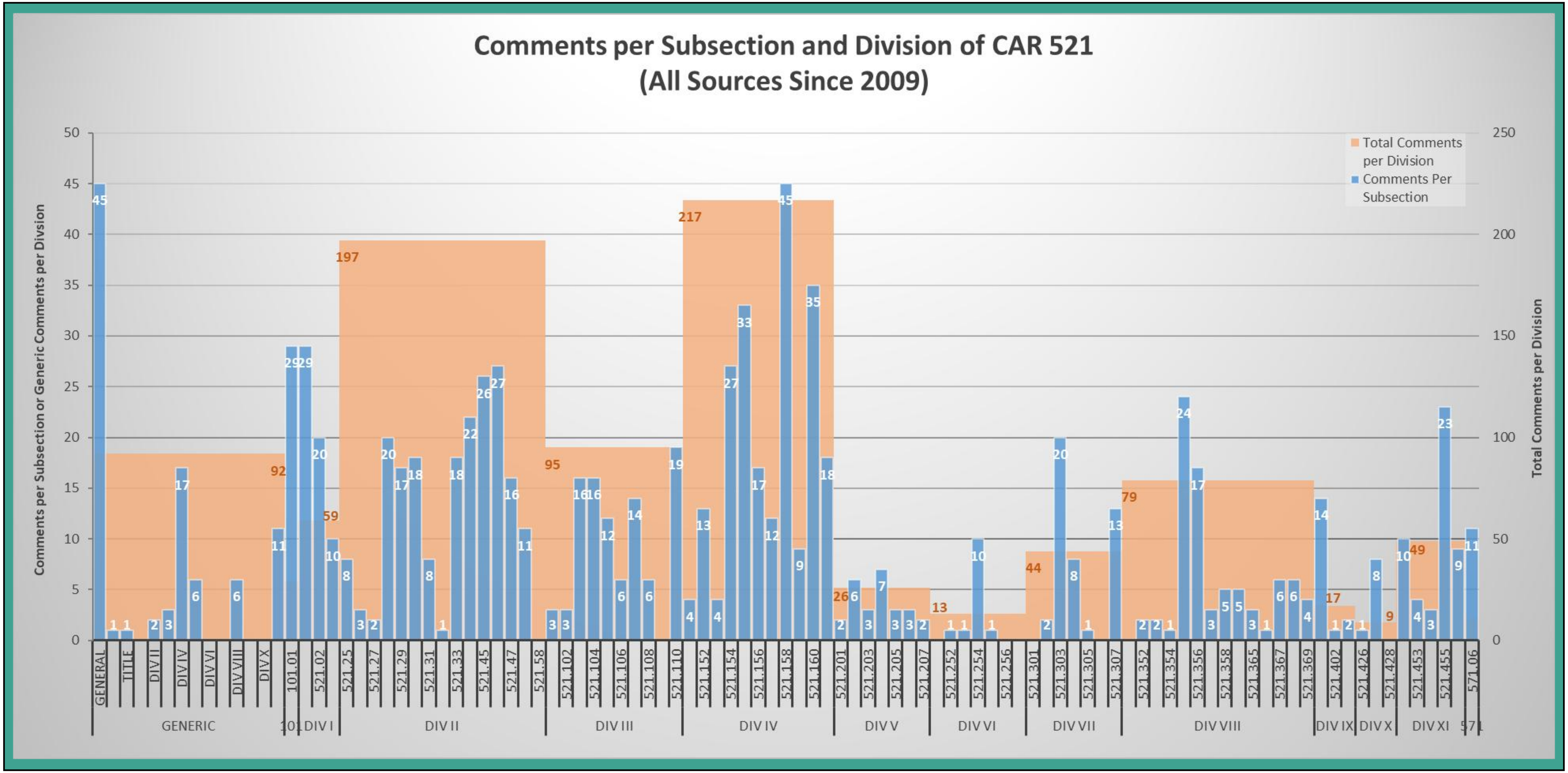


- Regulatory sandboxes are to be broadly enabled for Government of Canada regulators under amendments to the ***Red Tape Reduction Act***, for ministers otherwise lacking the legislative authority.
- The Minister of Transport has authority under 5.9(2) of the ***Aeronautics Act*** to exempt from the application of any regulation or order in consideration of a two-way test:
  - No adverse effect on aviation safety or security; and
  - In the public interest.
- The Minister of Transport has authority under 6.7(1) of the ***Canada Transportation Act*** to exempt, by order, for a period of not more than 5 years from any provision of legislation or an instrument made under that legislation that the Minister administers in consideration of a two-way test:
  - In the public interest; and
  - The exemption promotes innovation in transportation through research, development or testing.

CAR 521 Rewrite

Complete Rewrite of CAR 521 Underway:

- Aircraft Certification Standards (AARTC) is leading, engaging across Transport Canada, consulting with ‘volunteer’ stakeholders and seeking clarification from commenters.
- Scope of proposed amendments :
  - To address 732 comments recorded since CAR 521 came into force in 2009.
  - Considers comments on NPA 2010-021, NPA 2010-022, A/NPA 2014-001; Regulatory Review “irritants”; CAR 521 unsolicited feedback; House/Senate Standing Joint Committee for the Scrutiny of Regulations.
- TBS “significant impact”, requiring detailed Regulatory Impact Analysis Statement (RIAS):
  - Total industry cost would exceed 1 million in 1 year or 10 million in 10 years;
  - An evidence-based, non-technical synthesis of expected impacts, positive and negative, of a proposed regulation
- Current project timeline:
  - CARAC NPA (online for 75 days) – Target Begin Date: March 2027
  - Legal Drafting/RIAS – Target Begin Date: November 2027
  - Pre-Publication (Canada Gazette Pt.I) (online for 75 days) – Target Begin Date: January 2030
  - Publication (Canada Gazette Pt.II) – Target Registration Date: December 2030


Comments per Subsection and Division of CAR 521
(All Sources Since 2009)
Comments per Subsection or Generic Comments per Divsion
Total Comments per Division
Total Comments per Division
Comments Per Subsection
GENERIC
101
DIV I
DIV II
DIV III
DIV IV
DIV V
DIV VI
DIV VII
DIV VIII
DIV IX
DIV X
DIV XI
571

Regulatory Pressure Points

CAR 521.152(2)

- Prohibits a person to undertake a change to type design that is not a (major) change, except in accordance with CAR 521.154.

CAR 521.154

- Permits only the holder of a type design to make a change other than a (major) change, if made in accordance with procedures accepted by the Minister to ensure that the changed type design continues to comply with its existing certification basis.

CAR 571.06(1)(a)

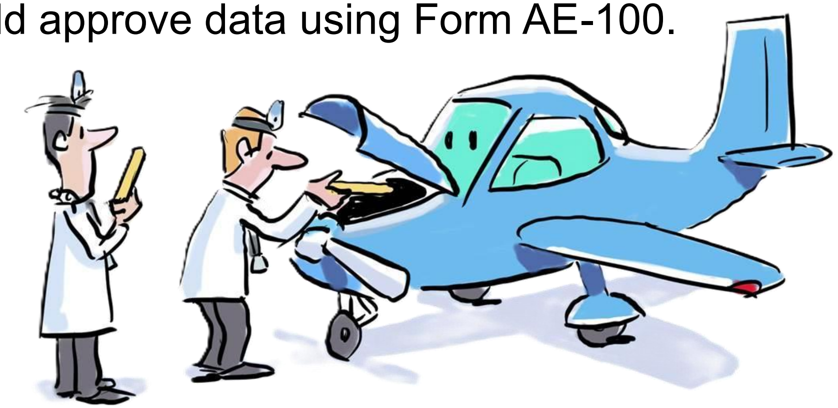
- A person who signs a maintenance release shall ensure that a “major modification” conforms to relevant and applicable “approved data”, which includes
  - design approvals under CAR 521; and
  - other drawings and methods approved by the Minister (but no enabling regulation to render this data).

CAR 571.10(1)

- Maintenance release - Declaration that the performance rules of CAR 571.02 have been complied with and the applicable standards of airworthiness have been met.

What are we trying to fix?

- In 1991, use of Form AE-100 for approving data (“one-off”) in support of major modifications was formally discontinued by Transport Canada due to misuse, leading to the current more stringent requirements.
- In 2004, stakeholder feedback at the CARAC Part V (Aircraft Certification) Technical Committee indicated that Transport Canada had ‘induced a major change industry’.
- In 2018, Government of Canada’s Regulatory Review recorded the Canadian aircraft modification industry assertion that it faces competitive disadvantages due to higher costs and longer approval times required under the current Canadian regulations as compared to U.S. competitors.
    - In the U.S., an FAA Designated Engineering Representative (DER) can approve data for some major alterations at a lower cost and shorter timeframe compared to the Canadian process, which requires an STC.
    - In Canada, a Design Approval Representative (DAR), Design Approval Organization (DAO), or Airworthiness Engineering Organization (AEO) issues a Form 26-0757 - Statement of Compliance, which does not approve data but confirms compliance with regulations. This contrasts with the previous practice where DAR/DAO/AEO could approve data using Form AE-100.



Current Industry Concerns

Regulatory Burden:

- The current design approval process requires a new 'full' STC to render approved data for modifications, even if the modification has a negligible effect on airworthiness or environmental characteristics. This incurs additional costs and time for both the design approval service provider and the aircraft owner/operator.

Threshold for "Major" Modifications:

- There is a perceived difference in the threshold for what constitutes a "major" modification between Canada and other jurisdictions like the U.S. and Europe. This discrepancy leads to additional work and costs for Canadian applicants.

Need for Streamlined Process:

- The industry has called for a more streamlined process that allows for the approval of data for simple major modifications without the need for an STC. This would help reduce costs, while improving the efficiency and competitiveness of the Canadian aircraft modification industry.

What fix are we going to try?

Provide a New Means to Render "Approved Data"

- Provide a means for the Minister to approve "other drawings and methods", meeting the definition of Standard 571.06(1) "approved data" (b).

Introduce Minor Changes to Type Design

- Prior to CAR 521 [SOR/2009-280, s. 26., 1 October 2009], regulations and standards (AWM, E&I Manual) referred to "major modifications" and "minor modifications", referring both to the alteration of an aeronautical product and to the design of the modification. The same term was used in both the maintenance context and the design context, leading to confusion.
- "Minor modification" terminology removed from maintenance regulatory framework in 1987 [AWM 571].
- Third-party (non-OEM) "minor modification" removed from aircraft certification regulatory framework in 1991 [AWM 513, First Ed.]. Now only referred to "design change", approved by supplemental type approval (STA).
- All references to "major" and "minor" removed from the regulatory framework for aircraft certification in 1998 [CARs 511, 513: SOR/98-526, 22 October 1998].
- CAR 521 now refers to "a change to the type design", which meets the criteria specified in CAR 521.152(1).
- Moving forward, will introduce classification of approved changes, as "minor change" and "major change".

Potential Benefits to Aviation Safety

Facilitate the introduction of 'safety-enhancing technologies' into the general aviation fleet.

Help further the goal of simultaneously enhancing safety and reducing burdens on industry, which was the aim of the "Part 23 Rewrite" [AWM 523 Change 523-18] that overhauled airworthiness standards for small general aviation (GA) airplanes.

By making it easier, faster, and less expensive to get safety-enhancing technologies into small airplanes, we will continue to reduce the number of fatal general aviation accidents and save lives.

Desired modifications include (among others):

- Non-required angle-of-attack indicators
- Moving map GPS
- Weather information systems
- Electronic conspicuity devices (ADS-B Out)
- Carbon monoxide detectors
- Non-required shoulder harnesses
  (e.g. front seats of small aeroplanes manufactured before July 18, 1978)

Angle-of Attack Indicator
(NORSEE)

Digital Panel Mounted CO Detector
(TSO-C48)

Targeting Safety Enhancements

The Number One and Number Two most fatal General Aviation accidents in Canada are:

1. Loss of Control – Inflight (47%)
2. Controlled Flight into Terrain (11%)

Related Transport Canada Efforts

A Streamlined Approach for Installation of Non-Required Safety Enhancing Equipment (NORSEE)

- **Exemption NCR-024-2022**
  - The exemption permits the eligibility under Subpart 571 of the CARs for installation of new NORSEE, marked with “14 CFR 21.8(d)”, on Canadian-registered aircraft with a Certificate of Airworthiness flight authority
  - NORSEE might be installed as an other than major modification.
  - In other cases, NORSEE installation might require design approval.

General Aviation Safety Program

- **See next slide**

General Aviation Safety Program (GASP)

Together with COPA, Transport Canada launched the General Aviation Safety Campaign in May 2017

Transitioned to GASP in June 2020

A series of GA Safety Seminars, most recently March 2025

Safety Priorities include:

- Better pilot decision making
    - Majority of GA fatalities in Canada are due to a Loss of Control In-flight (LOC-I).
    - Nearly 80% (est.) of all aviation accidents are related to human factors
- Staying current and proficient as a pilot
    - Keep your skills up to date with continual training
- Best practices
    - Passenger briefings
    - Checklists
    - Safety equipment

Past Civil Aviation focus on pilot education and awareness. However, better technology offers the pilot better safety tools!

Proposal

Regulatory Sandbox Proposal

- Transport Canada proposes to use a regulatory sandbox to explore and demonstrate a new “minor change approval” process for simple major modifications.
- Helps understand the impact on people, markets, and regulations.
- Seeks participants (interested volunteers) to trial test a new system.
- Temporary relaxation of regulatory constraints to permit testing of a new framework.
- Participants provide feedback to Transport Canada on their experiences in the sandbox.
- Aircraft Certification Standards (AARTC) will create a final evaluation report for the sandbox experience. The report will be made available publicly to share knowledge, insights, and outcomes transparently with stakeholders, including members of the public, other regulators, and government entities.

Outcome:

- Gather evidence to make informed decisions on possible amendments to CARs.

Implementation:

- Enabled by a global exemption from current regulations, NCR-028-2023.

We Will Need a New Language

Revised or New Definitions

- **Modification:** the performance of maintenance under CAR 571 on a type certificated aeronautical product that adds to and/or removes from the approved configuration; *primarily intended to refer to the physical alteration* done to a type certificated aeronautical product. The term "modification" should not be confused with the design of a modification (i.e. a change to an approved type design) under CAR 521.
- **Major modification:** an *alteration to an aeronautical product* in respect of which a type certificate has been issued that *might appreciably affect* weight, balance, structural strength, performance, power plant operation, flight characteristics, or other qualities affecting its airworthiness or environmental characteristics.
- **Change to a type design:** any change to *the set of data and information approved* under a type certificate (including an amended or supplemental type certificate) that is necessary to define an aircraft, aircraft engine, or propeller, together with its approved operating and airworthiness limitations, for the purpose of determining its airworthiness or environmental characteristics.
- **Minor change:** a change to a type design that *has no appreciable effect* on the weight, balance, structural strength, reliability, operational characteristics, or other qualities affecting the airworthiness or environmental characteristics of an aircraft, aircraft engine, or propeller.

Technical Data Required for Modifications

CAR 571

Other than Major Modification

Needs

Acceptable Data

From

- TCCA Advisories
- OEM Recommendations
- FAA AC 43.13-1B/-2B (or equivalent)
- ~~Data "found appropriate" by a delegate~~

CAR 571

Major Modification

Needs

Approved Data*

* "Specified data" is a form of approved data.

From

CAR 521

- Minor Change Approval
- STC
- Amended type certificate procedures

AMO/AME performs modification, entry made in technical record, maintenance release signed.

We Will Need Relaxed Regulations

Relaxed Regulations within the Regulatory Sandbox

- **CAR 521.152**
  - Relax current CAR 521.152 to include an additional option, to classify a change as a “minor change”.
  - This additional option would be in addition to the two existing options of either:

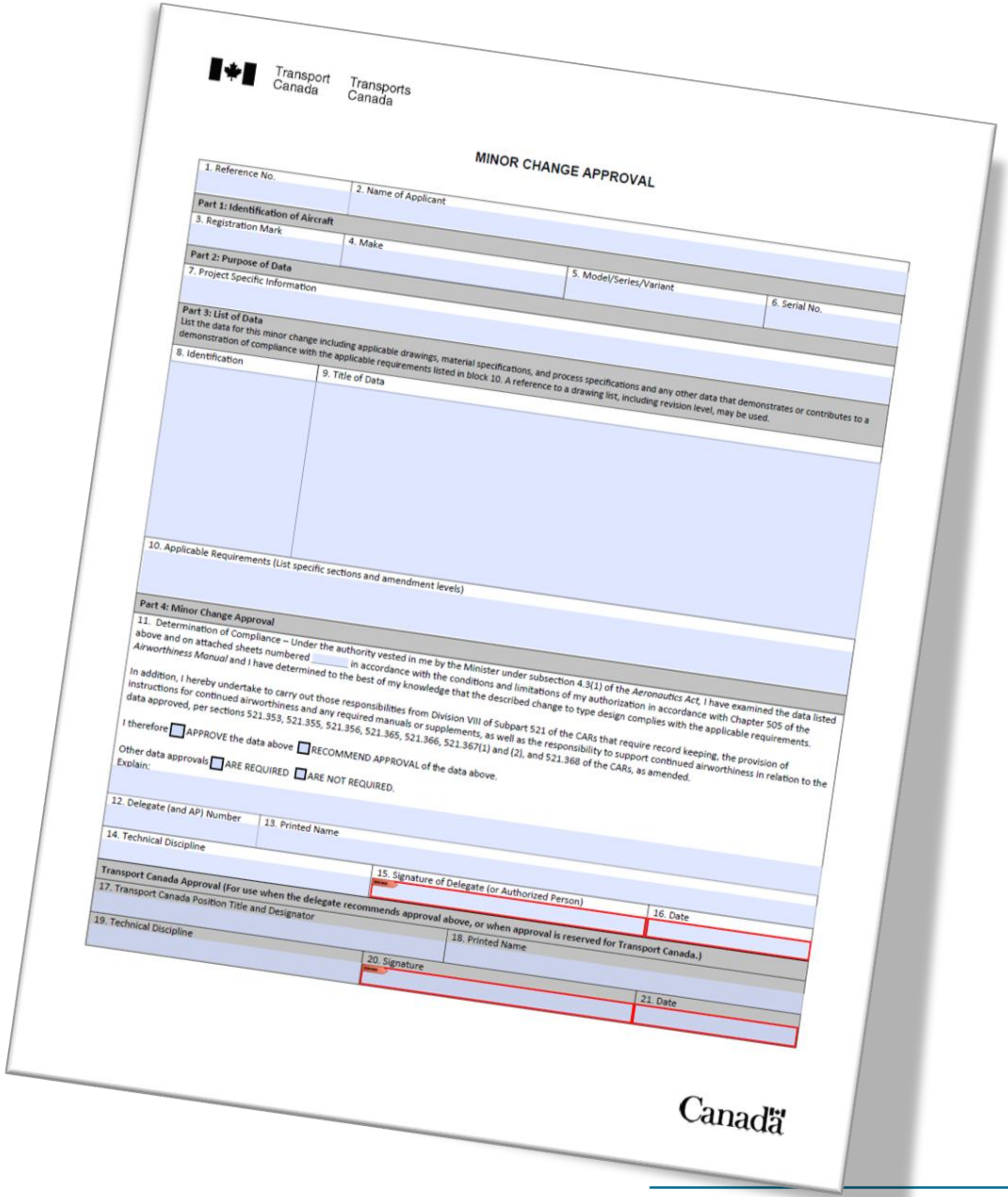
    (1) under CAR 521.152(1), a “change to the type design”; or

    (2) under CAR 521.152(2), a “change other than a change to the type design” referred to in 521.152(1).
- **CAR 571.06(1)(a) and 571.10(1)**
  - Partial relaxation in respect of the participation of an AMO/AME in the regulatory sandbox, to the extent that they would use relevant technical data approved by means of a minor change approval in support of the embodiment of an eligible major modification.
  - Provides regulatory cover for the use of a properly executed Minor Change Approval as “approved data”.
  - This relaxation may not strictly be necessary, but is included for completeness and for clarification, if required.

Minor Change Approval Form

Finding of Compliance and Approval:

- An authorized aircraft certification delegate uses their authority to
    - determine compliance of engineering data with the certification basis; and
    - to approve a minor change.
- A DAR or AP within an AEO or DAO signs and issues a Minor Change Approval form, approving the listed data.


Transport Canada Transports Canada

MINOR CHANGE APPROVAL

1. Reference No.
2. Name of Applicant

Part 1: Identification of Aircraft
3. Registration Mark
4. Make
5. Model/Series/Variant
6. Serial No.

Part 2: Purpose of Data
7. Project Specific Information

Part 3: List of Data
List the data for this minor change including applicable drawings, material specifications, and process specifications and any other data that demonstrates or contributes to a demonstration of compliance with the applicable requirements listed in block 10. A reference to a drawing list, including revision level, may be used.
8. Identification
9. Title of Data
10. Applicable Requirements (List specific sections and amendment levels)

Part 4: Minor Change Approval
11. Determination of Compliance – Under the authority vested in me by the Minister under subsection 4.3(1) of the *Aeronautics Act*, I have examined the data listed above and on attached sheets numbered ____ in accordance with the conditions and limitations of my authorization in accordance with Chapter 505 of the *Airworthiness Manual* and I have determined to the best of my knowledge that the described change to type design complies with the applicable requirements.

In addition, I hereby undertake to carry out those responsibilities from Division VIII of Subpart 521 of the CARs that require record keeping, the provision of instructions for continued airworthiness and any required manuals or supplements, as well as the responsibility to support continued airworthiness in relation to the data approved, per sections 521.353, 521.355, 521.356, 521.365, 521.366, 521.367(1) and (2), and 521.368 of the CARs, as amended.

I therefore ☐ APPROVE the data above ☐ RECOMMEND APPROVAL of the data above.

Other data approvals ☐ ARE REQUIRED ☐ ARE NOT REQUIRED.

Explain:

12. Delegate (and AP) Number
13. Printed Name
14. Technical Discipline
15. Signature of Delegate (or Authorized Person)
16. Date

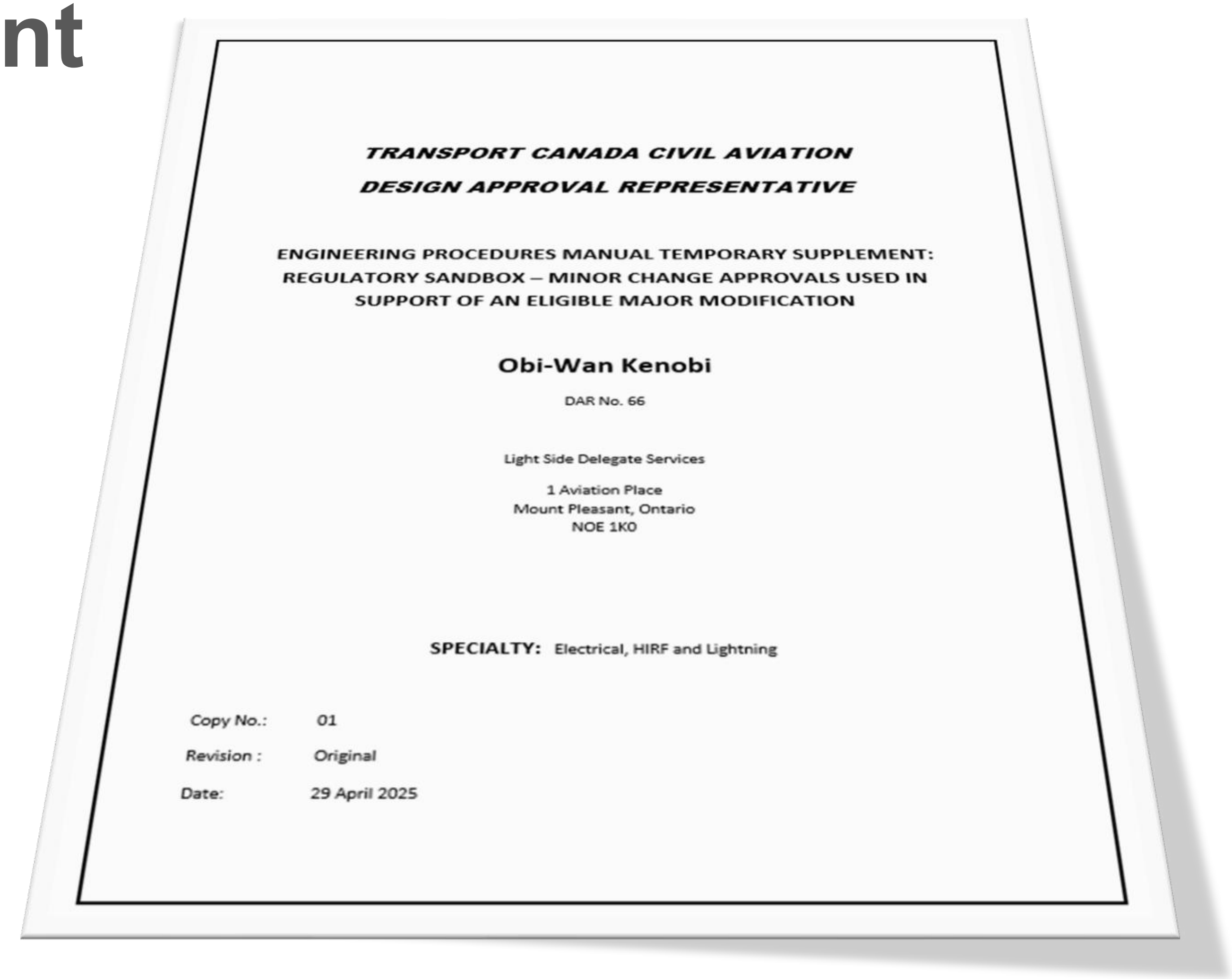
Transport Canada Approval (For use when the delegate recommends approval above, or when approval is reserved for Transport Canada.)
17. Transport Canada Position Title and Designator
18. Printed Name
19. Technical Discipline
20. Signature
21. Date

Canada

Temporary Supplement to EPM or DAPM

As a condition of use of Exemption NCR-028-2023 by an aircraft certification delegate that has been selected by Transport Canada to participate in the regulatory sandbox, the aircraft certification delegate must

- develop and obtain Transport Canada approval of a minor change approval temporary supplement to their approved delegate procedures manual.
- detail the procedures that the aircraft certification delegate will use to review and approve minor changes.


TRANSPORT CANADA CIVIL AVIATION
DESIGN APPROVAL REPRESENTATIVE

ENGINEERING PROCEDURES MANUAL TEMPORARY SUPPLEMENT: REGULATORY SANDBOX – MINOR CHANGE APPROVALS USED IN SUPPORT OF AN ELIGIBLE MAJOR MODIFICATION

Obi-Wan Kenobi

DAR No. 66

Light Side Delegate Services

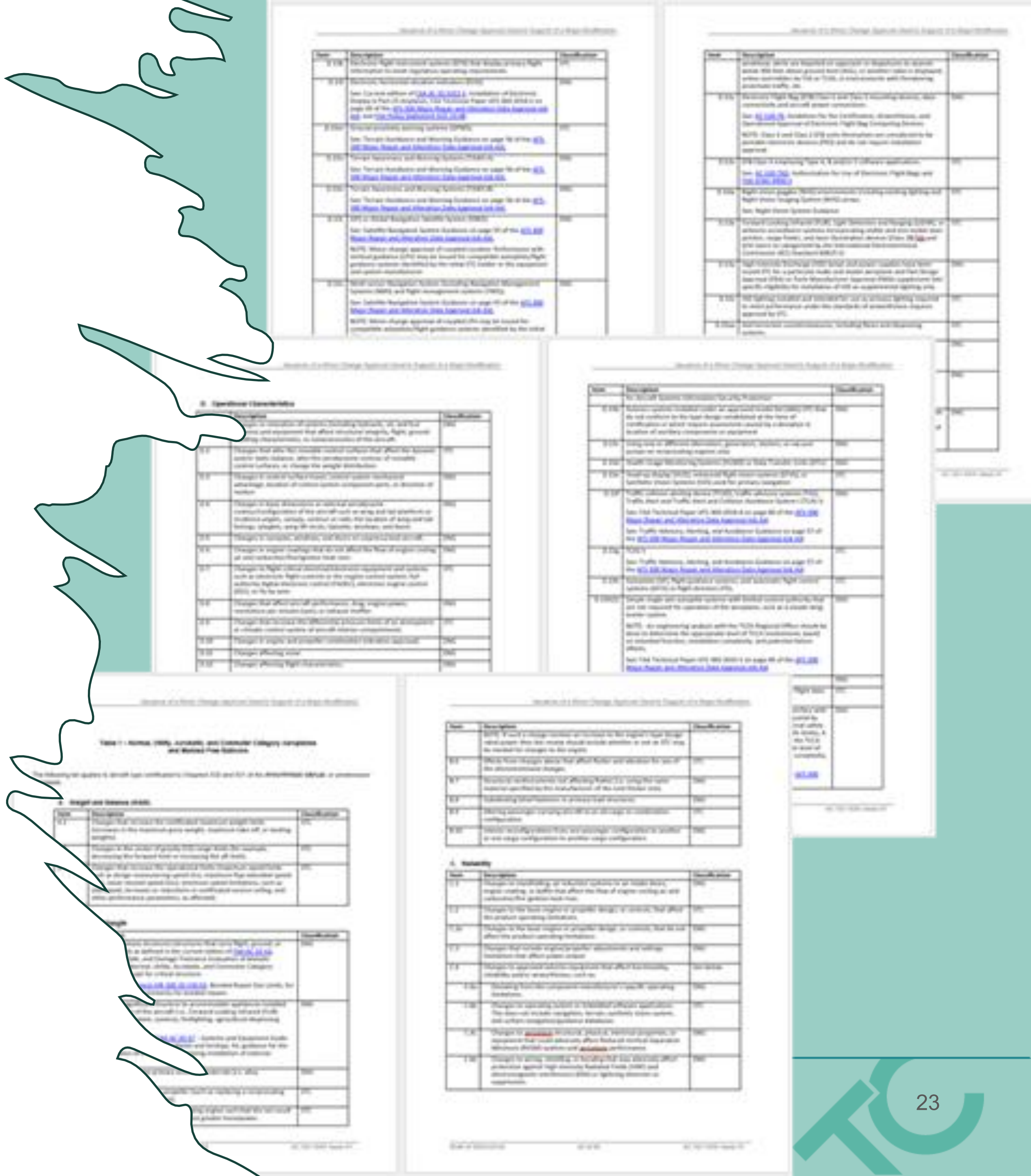
1 Aviation Place
Mount Pleasant, Ontario
NOE 1K0

SPECIALTY: Electrical, HIRF and Lightning

Copy No.: 01
Revision : Original
Date: 29 April 2025

How Do I Know if I Can Classify the Change as “Minor”?

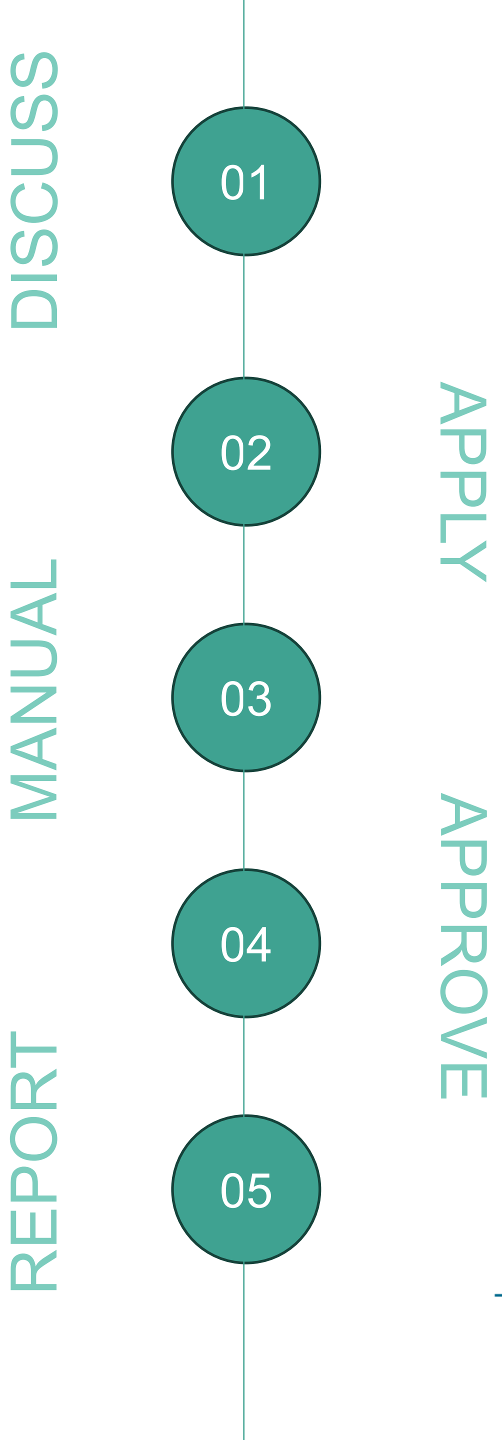
- The change must be **within my scope** of authority.
- I must determine that the change, when embodied, would **have no appreciable effect** on the weight, balance, structural strength, reliability, operational characteristics, or other qualities affecting the airworthiness or environmental characteristics of an aircraft, aircraft engine, or propeller.
- The proposed major modification must be **eligible** for approval as a minor change. [See Appendix A to draft AC 521-XXX]



Process

Participation in the regulatory sandbox is VOLUNTARY:

- There is no obligation to participate.
- Transport Canada does not compensate participants.
- Participants understand that the minor change approval process is EXPERIMENTAL and that there may be deficiencies in guidance and/or advisory material to support necessary processes.
- Participants are expected to provide feedback to Transport Canada on their experience with the sandbox processes, such as:
    - Comments on possible process improvements
    - Comments on test advisory and guidance materials
    - Assessment of costs/benefits as it applied to their projects, including for the AMO/AME and aircraft owner



Regulatory Sandbox Procedures

(1) Eligibility

- Any natural or legal person whose aeronautical engineering qualifications have been recognized through having been authorized pursuant to section 4.3(1) of the *Aeronautics Act* as a:
  - Design Approval Representative (DAR);
  - Design Approval Organization (DAO); or
  - Airworthiness Engineering Organization (AEO).
- Limited to third-party (non-design approval holder) aircraft certification design service providers.
- Have a clearly developed plan for live environment testing:
  - provide a business model (i.e., anticipated projects (e.g. aircraft types, modifications, geographic region, customers (private, corporate, single or fleet owner));
  - identify the anticipated engineering discipline(s) in their scope; and
  - explain what they think the impacts of using the minor change approval process might be.
- DAR/DAO/AEO shall commit to provide regular feedback (at least each 6 months) to TCCA .

Regulatory Sandbox Procedures

(2) Application to Participate:

- Assess your eligibility
- Discuss with your RMAC or Delegations Division (AARDL) in HQ
- Apply by letter
- Be selected by TCCA to participate.

(3) Authorization to Issue Minor Change Approvals:

- Transport Canada will approve an EPM/DAPM Temporary Supplement; and
- Transport Canada will provide a Letter of Authorization for Minor Change Approvals.

(4) Accomplishment and Report Results:

- For work accomplished by AMO/AME on aircraft supported by a Minor Change Approval, record your assessment of how using the minor change approval process impacted your work and the overall aircraft owner’s project.

Regulatory Sandbox Procedures

(5) Account for your activity:

- In this context, a minor change approval **is considered a design approval**.
- A minor change approval is to be **recorded in NAPA**.
- A minor change approval **is treated differently than a major change** as it pertains to the assignment of design approval holder (DAH) responsibilities.
  - **In the case of a major change**, the successful applicant for a major change becomes the DAH for that change upon its approval. A DAH in respect of a major change to a type design has responsibility for the approved design change.
  - **In the case of a minor change**, the applicant is typically not able to support the design of an aeronautical product. They are referred to as the ‘applicant’ only as a convention in the context of a minor change approval process, but it is not intended that they become the holder.
- In respect of a minor change approval, **the aircraft certification delegate assumes certain holder responsibilities** (but not all) from Division VIII of CAR 521. In particular:
  - Maintain and provide access to the Minister the technical data used to approve the minor change;
  - Provide instructions for continued airworthiness (ICA) and any required manuals or supplements; and
  - Be responsible to support continued airworthiness in relation to the data approved.

Exit Strategy for Participants

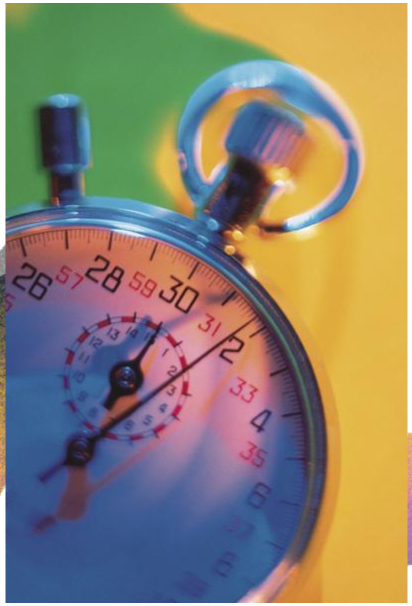
EPM / DAPM Temporary Supplements

- Authorizations to DAR/DAO/AEO to approve minor changes **will expire after 5 years**, and procedures contained in EPM/DAPM temporary supplements may no longer be used.
- For this reason, the **main body of an EPM/DAPM should not be revised** to include procedures for the regulatory sandbox so that use of the EPM/DAPM Temporary Supplement may simply be discontinued without consequence to the pre-existing procedures.

Transitions and Future Directions

- If the minor change approval process tested may be considered 'market ready', it will be evidenced by the readiness with which questions around regulatory compliance may answered based on the outcomes of testing.
- Regulatory amendments will be proposed in consideration of regulatory sandbox results.
- Transition measures to future proposed regulatory requirements might be set out for participants (where applicable), including pathways to become fully authorized. For example, TCCA may grant or issue:
    - subsequent exemptive relief from regulations to permit a continuation of a DAR/DAO/AEO's processes, as evolved and agreed to with TCCA.
    - a statement of regulatory compliance or comfort.

Key Takeaways



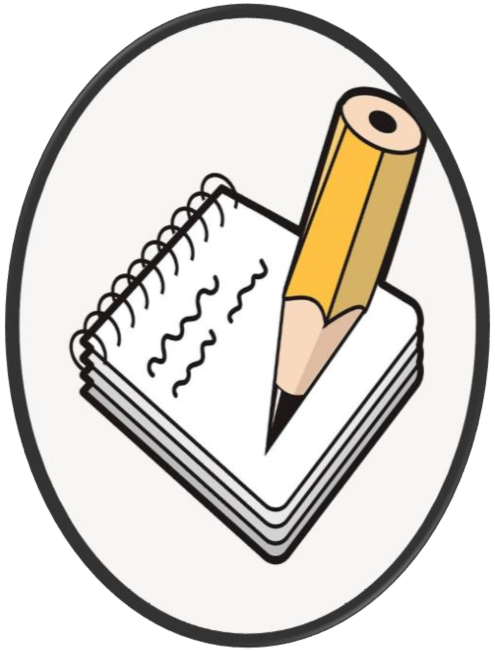
Regulatory sandboxes are a temporary, time-bound relaxation of regulatory requirements.

Minor changes to type design are not the same as **minor modifications** to an aircraft.
Major modifications may constitute either a minor change or a major change.

Application to participate
Authorization from TCCA
Approve minor changes
Accomplishment! Report results
Account for your activity

Questions ?

Thank you for your attention and interest today



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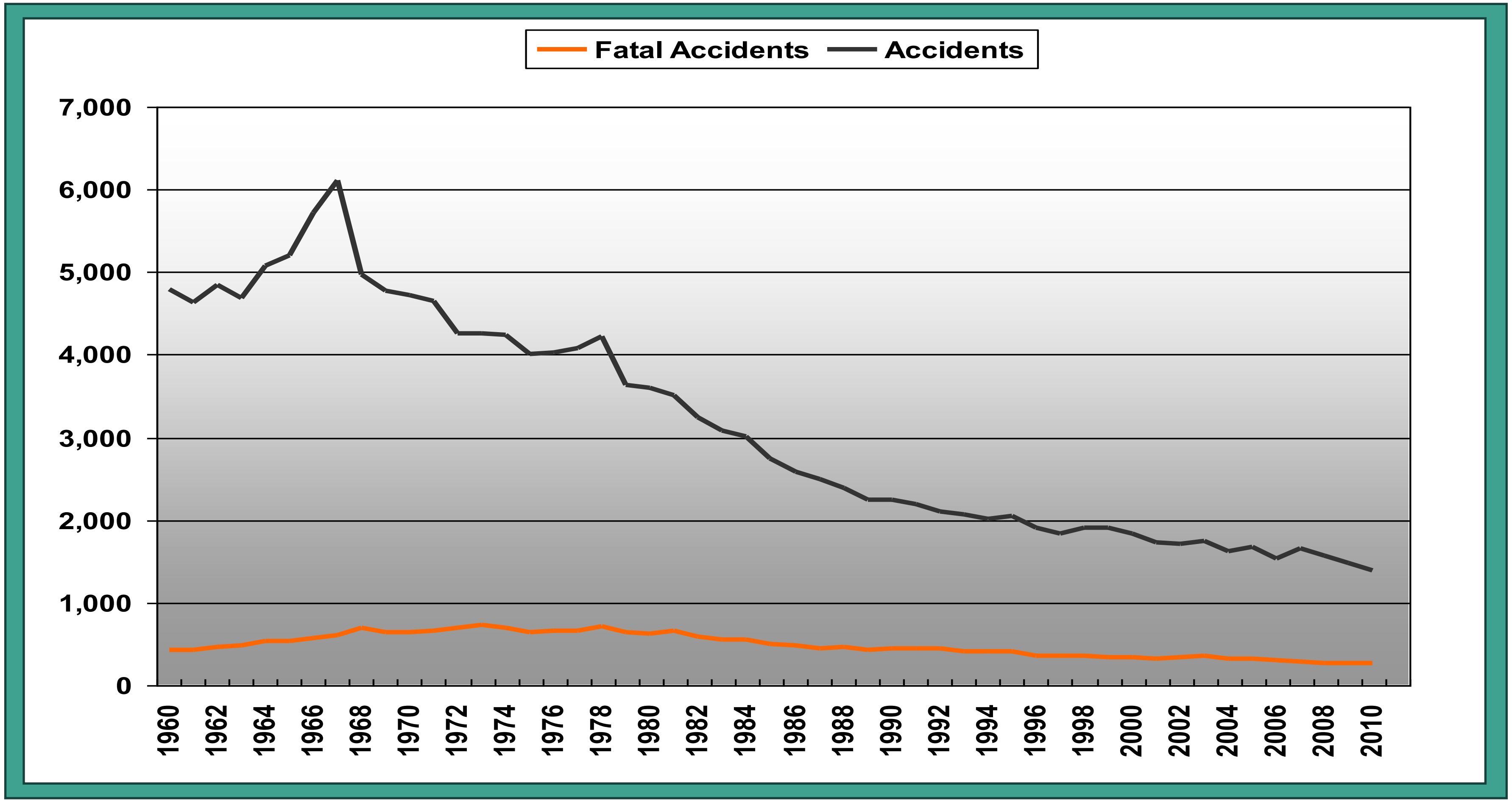
Extra Slides

General Aviation Accidents



FATAL GA ACCIDENTS

NTSB/GA-JSC 2008-2015